

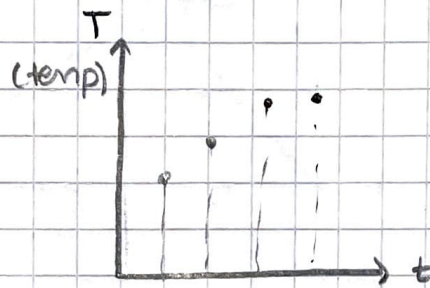
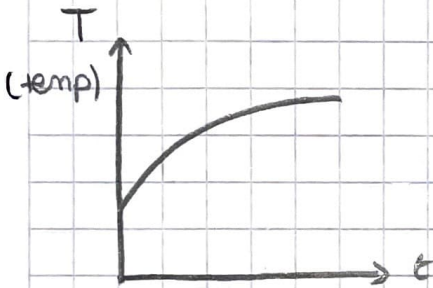
Logic Circuits

Definitions
Digital
Logic
Gate

Digital and Analog Quantities

Logic Level

- Activation Level
- Positive and Negative Logic



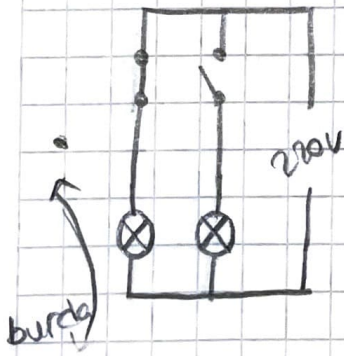
→ Zaman bakımından sürekli olan değerlere analog değerler

→ örneklenmiş, ayrışmış, kuantalanmış ise ⇒ digital (nicelendirilmiş)

Logic Levels

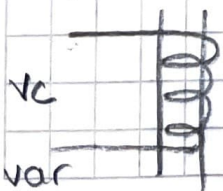
(H) High = 1 = Set = True = High Voltage (5V)

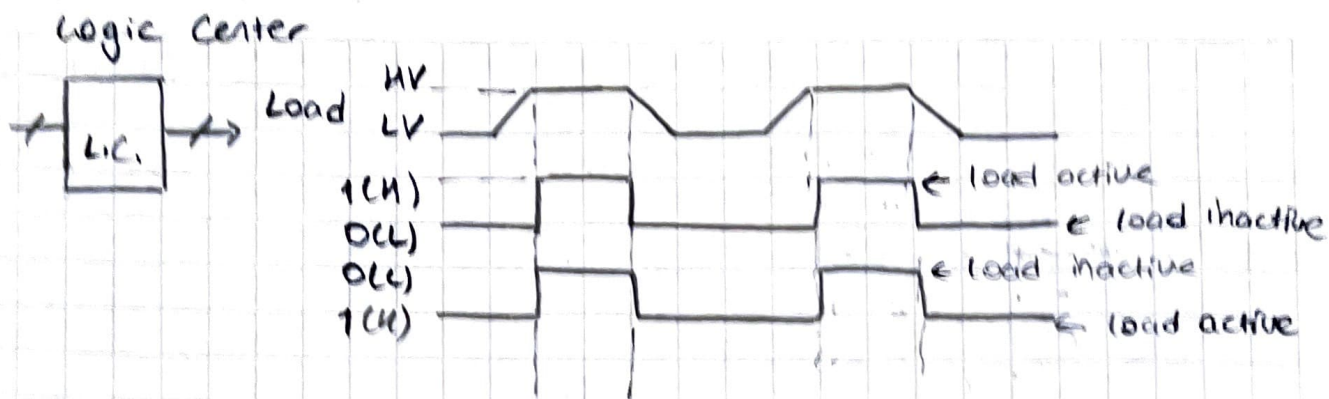
(L) Low = 0 = Reset = False = Low Voltage (0V)



$V_{\text{control}} = 0$ $L_1 = \text{ON}$ $L_2 = \text{OFF}$

$V_{\text{control}} = 5V$ $L_1 = \text{OFF}$ $L_2 = \text{ON}$





Logic Levels

HV $\leftrightarrow$ Logic 1

LV $\leftrightarrow$ Logic 0

} Positive Logic

HV $\leftrightarrow$ Logic 0

LV $\leftrightarrow$ Logic 1

} Negative Logic

Positive Logic



positive potential sources



negative potential sources

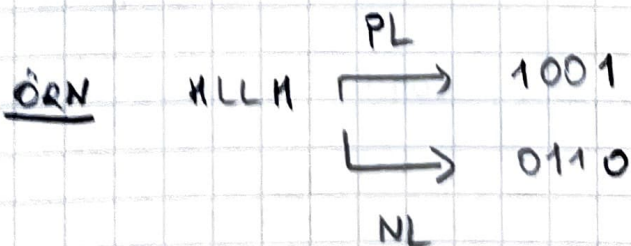
Negative Logic



positive potential sources

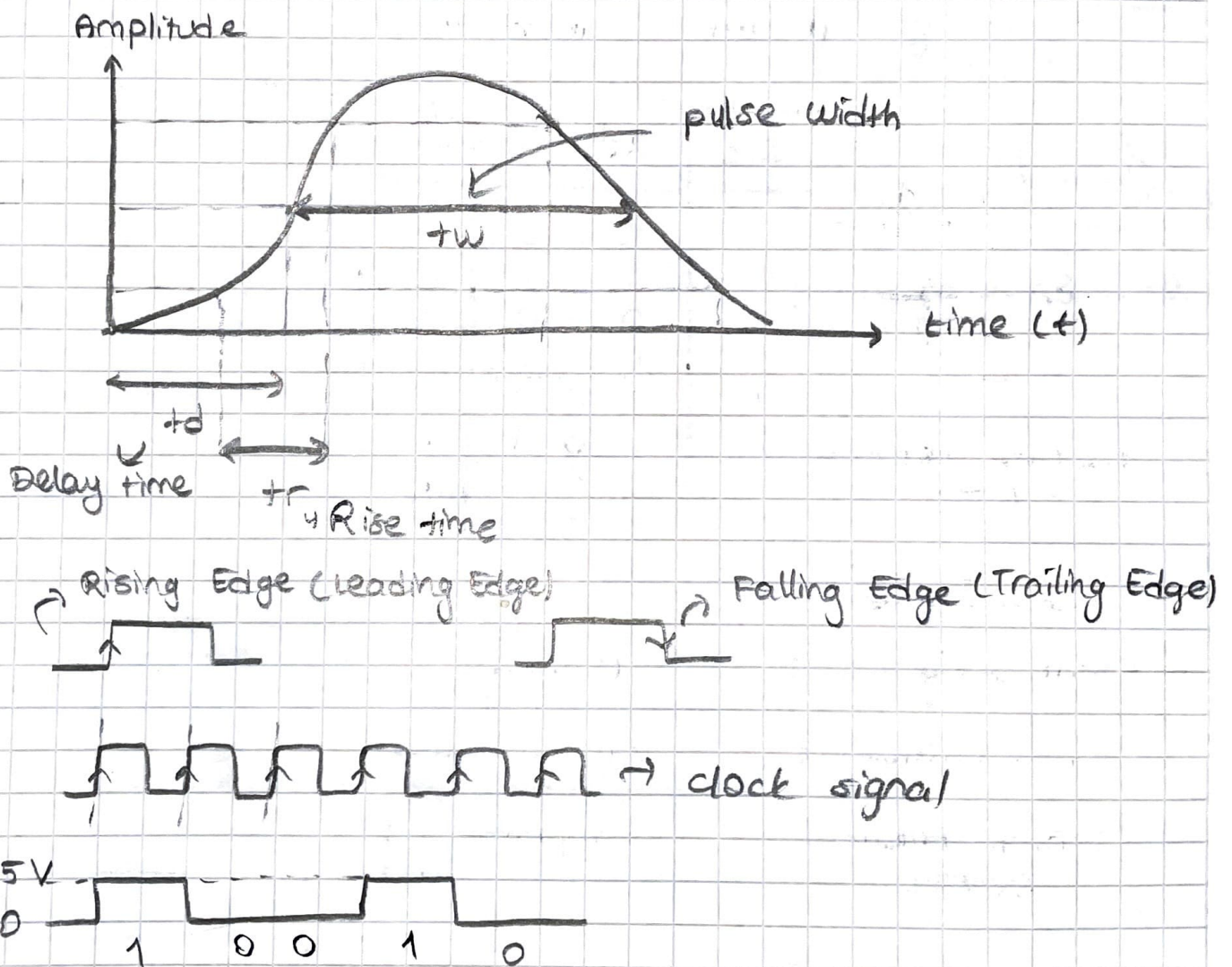


negative potential sources

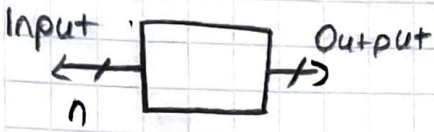


	<u>TTL</u>	<u>CMOS</u>
HIGH	-5V	5V
	-2.10V	3.5V
UNCERTAIN	-0.8V	1.5V
LOW	0V	0V

Pulse Waveform



Truth Table : Bir sistemin girişleri ile çıkışları arasındaki davranışlar.



n tane giriş $\Rightarrow 2^n$ tane durum

Decimal	Input		Output
	A	B	f
0	0	0	1
1	0	1	0
2	1	0	0
3	1	1	1

ÖRN Bir trafik lambasında iki veya daha fazla bnaa aynı anda yandığı zaman alarm veren devrenin doğruluk tablosunu çiziniz.



$2^3 = 8$ durum

Dec	R	Y	G	F
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	1
7	1	1	1	1

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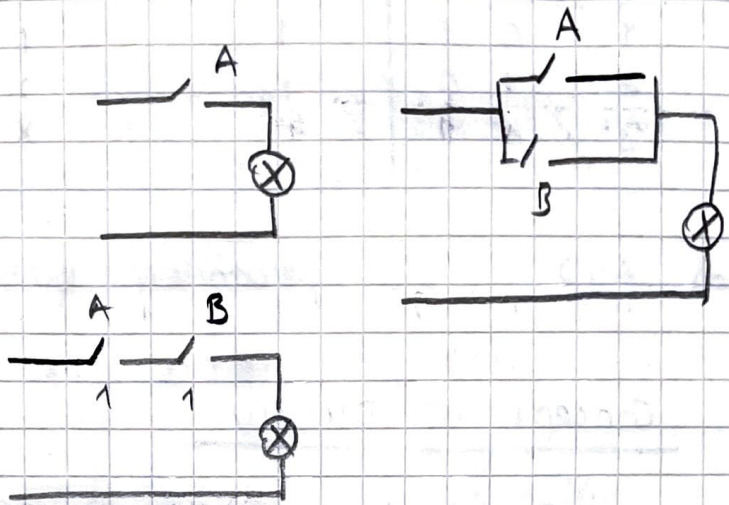
LAWS OF BOOLEAN ALGEBRA

Operators

NOT (-, ')

AND (·)

OR (+)

NOT OPERATOR (Complementation)

Dec	x	f = $\bar{x}$
0	0	1
1	1	0


 $\Rightarrow \bar{\bar{x}} = x \rightarrow \text{Involution (Erkennnis)}$
AND OPERATOR

Dec	x	y	f = $x \cdot y$
0	0	0	0
1	0	1	0
2	1	0	0
3	1	1	1

AND LAWS $x \cdot 0 = 0$ (Absorber) $x \cdot 1 = x$ (Neutral, Dummy) $x \cdot x = x$ $x \cdot \bar{x} = 0$

OR OPERATOR

Dec	X	Y	$f = x + y$
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	1

OR LAWS

$$0 + x = x$$

$$1 + x = 1$$

$$x + x = x$$

$$x + \bar{x} = 1$$

$\Rightarrow$ AND ve OR operatörleri birbirlerinin DUAL'idir.

Concept of Duality

$$0 \Leftrightarrow 1$$

AND ve OR operatörleri birbirinin dualidir.

$$(\cdot) \Leftrightarrow (+)$$

$\Rightarrow$ OR operatöründe girişlerden biri 1 ise sonuç 1
" " girişlerden biri 0 ise sonuç x

Duality Örnek

$$1 + x = x \Leftrightarrow x \cdot 0 = 0 \Rightarrow x + 1 = 1$$

LAWS OF BOOLEAN ALGEBRA

Associative Law (Birleşme Yasası)

Commutative Law (Değişim ")

Distributive Law (Dağıtma ")

Absorptive Law (Yutma ")

ASSOCIATIVE LAW

$\forall x, y, z \in B$ (Boole set)

$$(x \cdot y) \cdot z = x \cdot (y \cdot z) = x \cdot y \cdot z$$

$$(x + y) + z = x + (y + z) = x + y + z$$

COMMUTATIVE LAW

$$x \cdot y \cdot z = y \cdot x \cdot z = y \cdot z \cdot x$$

$$x + y + z = y + x + z = y + z + x$$

DISTRIBUTIVE LAW

$$x \cdot (y + z) = x \cdot y + x \cdot z$$

$$x + (y \cdot z) = (x + y) \cdot (x + z)$$

ABSORPTIVE LAW

$$x \cdot (\bar{x} + y) = x \cdot y$$

$$x + (\bar{x} \cdot y) = x + y$$

INVOLUTION

$$\overline{\overline{x}} = x$$

DE MORGAN LAW

$$\overline{x \cdot y} = \bar{x} + \bar{y}$$

$$\overline{x + y} = \bar{x} \cdot \bar{y}$$

NEUTR ELEMENT

$$x + 0 = x$$

$$x \cdot 1 = x$$

ABSORPTION ELEMENT

$$x \cdot 0 = 0$$

$$x + 1 = 1$$

IDEMPOTENTS

(Eş kuvvet)

$$x \cdot x = x$$

$$x + x = x$$

Cok
dremli

→ XNOR, XOR
dhenli

LAWS OF BOOLEAN ALGEBRA

Exclusive OR (EXOR, XOR) $\oplus$

Equivalence = Exclusive NOR $\odot$ (XNOR, EXNOR)

Dec	x	y	$x \oplus y$	$x \odot y$
0	0	0	0	1
1	0	1	1	0
2	1	0	1	0
3	1	1	0	1

$$* \quad x \oplus y = x \cdot \bar{y} + \bar{x} \cdot y = \overline{x \odot y}$$

$$* \quad x \odot y = x \cdot y + \bar{x} \cdot \bar{y} = \overline{x \oplus y}$$

EXOR

EXNOR

$$\left. \begin{aligned} x \oplus 0 &= x \\ x \oplus 1 &= \bar{x} \\ x \oplus x &= 0 \\ x \oplus \bar{x} &= 1 \end{aligned} \right\}$$

Duality

$\Leftrightarrow$

$$\left. \begin{aligned} x \odot 1 &= x \\ x \odot 0 &= \bar{x} \\ x \odot x &= 1 \\ x \odot \bar{x} &= 0 \end{aligned} \right\}$$

ASSOCIATIVE LAW

$$(x \odot y) \odot z = x \odot (y \odot z)$$

$$(x \oplus y) \oplus z = x \oplus (y \oplus z)$$

COMMUTATIVE LAW

$$x \odot y \odot z = x \odot z \odot y = z \odot x \odot y$$

$$x \oplus y \oplus z = x \oplus z \oplus y = z \oplus x \oplus y$$

ABSORPTIVE LAW

$$x \cdot (\bar{x} \oplus y) = x \cdot y \Rightarrow (x \cdot \bar{x}) \oplus (x \cdot y) = x \cdot y$$

$$x + (\bar{x} \odot y) = x + y \Rightarrow (x + \bar{x}) \odot (x + y) = x + y$$

DISTRIBUTIVE LAW

$$x \cdot (y \oplus z) = (x \cdot y) \oplus (x \cdot z)$$

$$x + (y \odot z) = (x + y) \odot (x + z)$$

DE MORGAN LAW

$$\overline{x \oplus y} = \bar{x} \odot \bar{y} = x \odot y \quad ?$$

$$\overline{x \odot y} = \bar{x} \oplus \bar{y} = x \oplus y \quad ?$$

→ EXOR : Girişlerden yalnızca 1 tanesi 1 old. 1 çıkış olur.
⊕ Fark dedektörüdür
Girişler birbirinden farklıysa çıkış 1

→ XNOR, EXNOR : Girişler aynıysa 1
⊙ Eşitlik dedektörü

LOGIC GATES

NOT Circuit = Inverter

AND Gate

OR "

NOR "

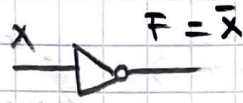
NAND "

EXOR "

XOR "

NOT CIRCUIT = INVERTER

X	F
0	1
1	0



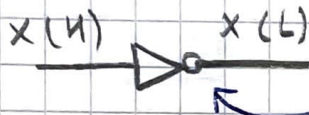
Active High to Active Low

Active Low to Active High

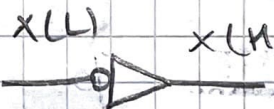
X	F = $\bar{X}$
LV	HV
HV	LV

X (L)	X (H)
1	1
0	0

X (H)	X (L)
0	0
1	1



Active Low Indicator (Bubble)

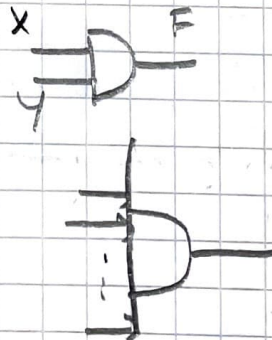


Positif lojikten negatif lojik'e çeviriyor.

Negatif lojikten positif lojik'e çeviriyor.

AND GATE

Dec	X	Y	F = X.Y
0	0	0	0
1	0	1	0
2	1	0	0
3	1	1	1

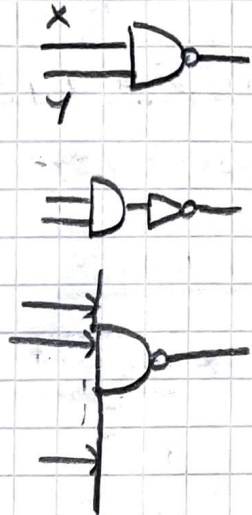


Positive Logic - Negative Logic

$$\left[\begin{array}{l} HV \leftrightarrow 1(H) = 0(L) \\ LV \leftrightarrow 0(H) = 1(L) \end{array} \right]$$

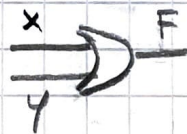
NAND GATE

Dec.	x	y	$f = x \cdot y$	$f' = \overline{x \cdot y}$
0	0	0	0	1
1	0	1	0	1
2	1	0	0	1
3	1	1	1	0



OR GATE

Dec	x	y	$F = x + y$
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	1



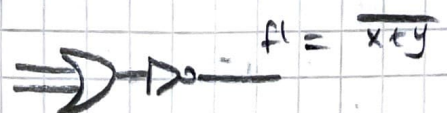
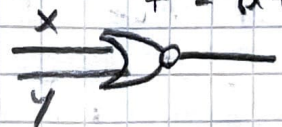
NOR GATE

Dec	x	y	$F = x + y$	$F' = \overline{x + y}$
0	0	0	0	1
1	0	1	1	0
2	1	0	1	0
3	1	1	1	0

$$(\bar{x} \cdot \bar{y})$$

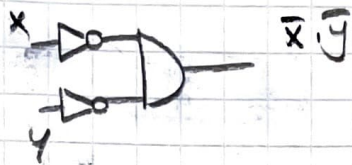


$$f' = \overline{x + y}$$



ÖRN

$$f' = (x+y) = \bar{x} \cdot \bar{y}$$



UNIVERSALITY OF NAND GATES and OR GATES

NAND GATE IMPLEMENTATION

INVERTER

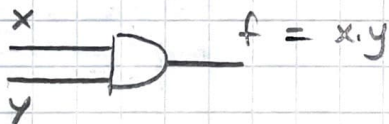


$$F = \overline{x \cdot y} = \overline{x \cdot x} = \bar{x}$$

y yerine x koyarsak



AND GATE

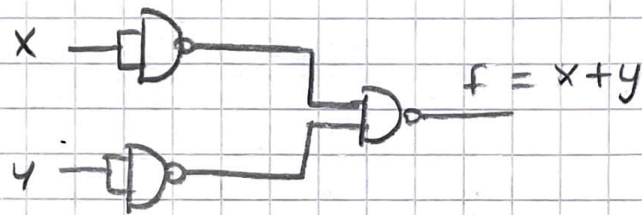


$$\bar{f} = \overline{\overline{x \cdot y}} \text{ } \{ \text{NAND (Inverter olan)} \}$$



OR GATE

x
 y  $f = x + y \Rightarrow \bar{f} = \overline{x + y} = (\bar{x} \cdot \bar{y})$ } NAND
NAND Inverter NAND Inverter
 $f_1 = \overline{x \cdot y}$
 $= \overline{A \cdot B}$



NOR GATE

INVERTER

x  $f = \bar{x}$

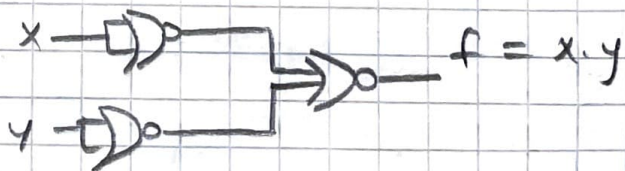
x
 y  $f = \overline{x + y} = f = \overline{x + x} = \bar{x}$

x  $f = \bar{x}$

AND GATE

x
 y  $\bar{f} = \overline{x \cdot y} = (\bar{x} + \bar{y})$ } NOR
NOR NOR

x
 y  $f_1 = \overline{x + y}$



OR GATE

$$x \Rightarrow y \Rightarrow f = x + y = \overline{\overline{x+y}} \quad \text{NOR (Inverter)} \quad \text{NOR}$$

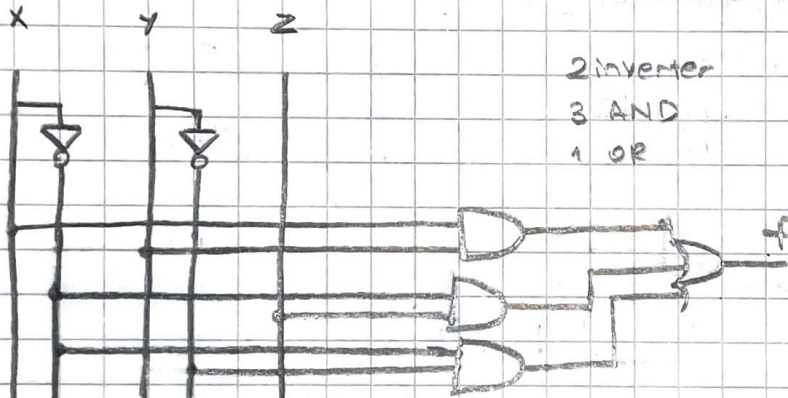
$$x \Rightarrow y \Rightarrow f = x + y$$

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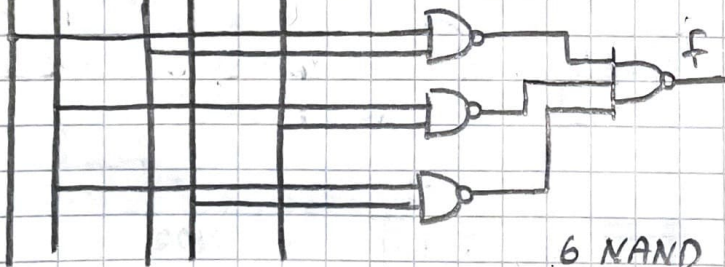
ÖRN $f = x.y + \bar{x}.z + \bar{x}.\bar{y}$ a) Temel lojik kapılarla gerçekleştirin.

b) Bu fonk. yalnızca NAND kapıları kullanarak gerçek.

$f(x,y,z)$



2 inverter
3 AND
1 OR



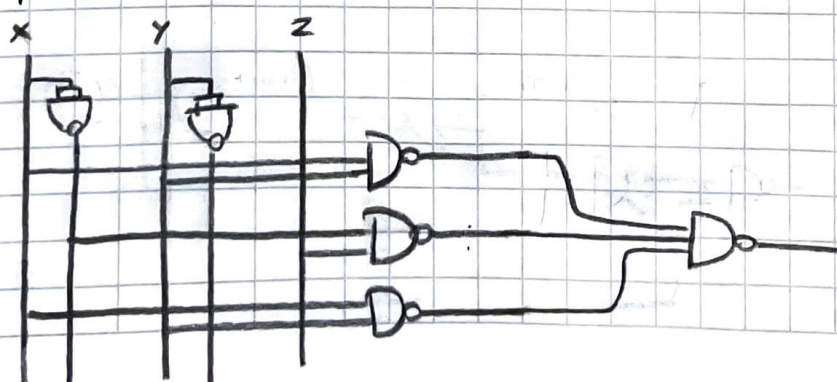
b) $f = x.y + \bar{x}.z + \bar{x}.\bar{y}$

$$\bar{f} = \overline{x.y + \bar{x}.z + \bar{x}.\bar{y}}$$

$$f = \overline{\overline{x.y} . \overline{\bar{x}.z} . \overline{\bar{x}.\bar{y}}}$$

6 NAND

$$x \Rightarrow y \Rightarrow f = \overline{x.y}$$



EXOR = XOR GATE

Dec	x	y	f = x ⊕ y
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	0

$$f = x\bar{y} + \bar{x}y = x \oplus y$$



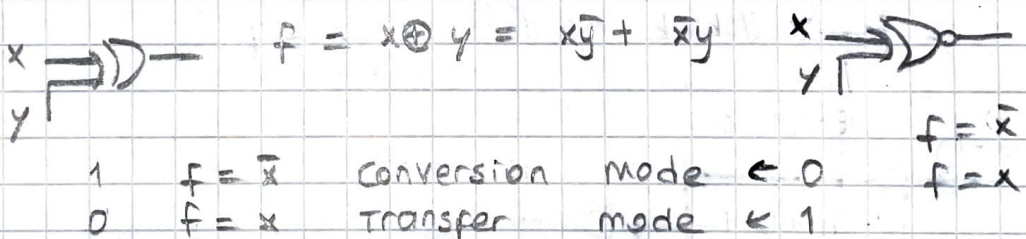
EQUIVALENCE = EXNOR = EQV

Dec	x	y	f = x ⊙ y
0	0	0	1
1	0	1	0
2	1	0	0
3	1	1	1

$$f = xy + \bar{x}\bar{y} = x \odot y$$



CONTROLLED LOGIC LEVEL CONVERSION



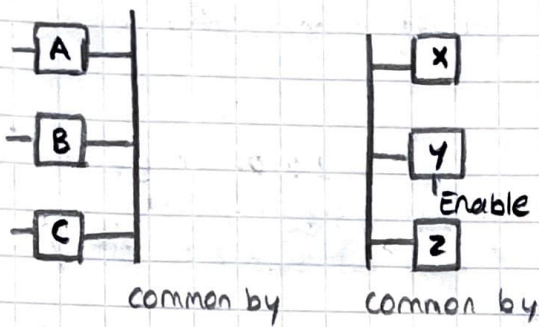
BUFFER (Tampon)

x	y	f = x
0	0	0
1	1	1



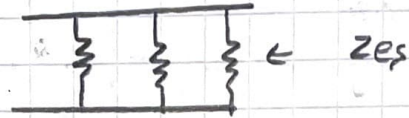
→ Sinyalin seviyesini deęistirmeden ideal hale getirin

TRISTATE GATES (3 durumlu kapılar)

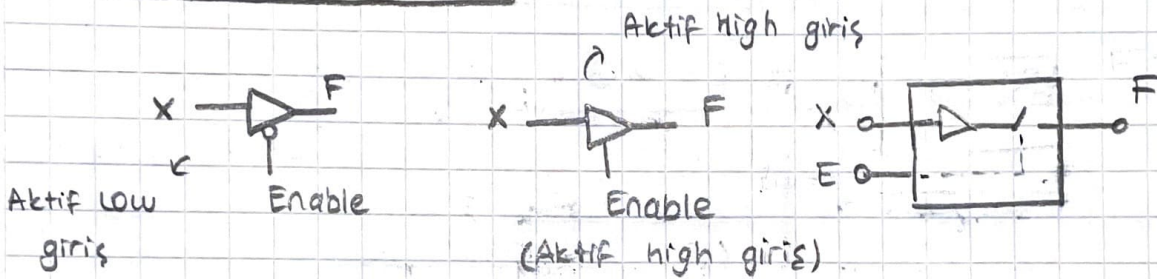


$Z \rightarrow$ empedans

$$\frac{1}{Z_{es}} = \frac{1}{Z_A} + \frac{1}{Z_B} + \frac{1}{Z_C}$$



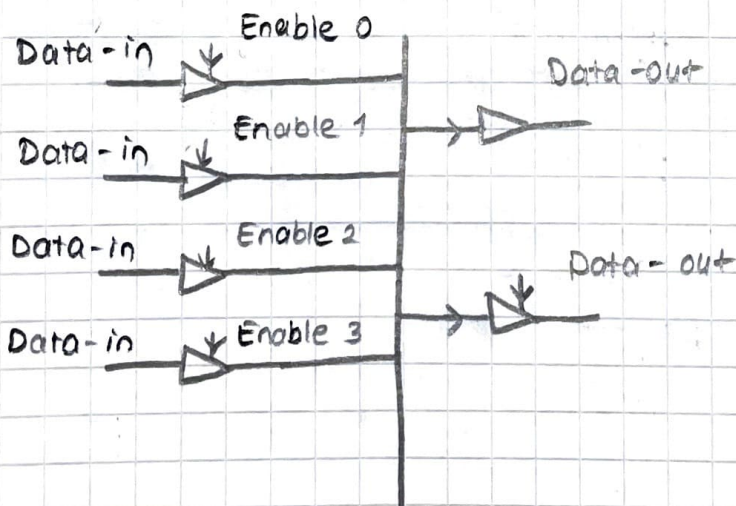
→ TRISTATE BUFFER



$\bar{E}$	F
0	x
1	Hi-Z

E	F
0	Hi-Z
1	x

↳ yüksek empedans



Tri-state Digital Buffer Data Bus Control

CONJUGATE GATE SYMBOL (ALTERNATE GATE SYMBOL)

De Morgan

$$(\overline{x \cdot y}) = (\bar{x} + \bar{y})$$

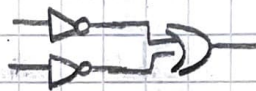
(L) ifadesinden

$$(\overline{x \cdot y}) \text{ anlamı } \leftarrow (x \cdot y) \text{ (L) } = (\bar{x} + \bar{y}) \text{ (H)}$$

geliyor.

1. Operatör değiştiriliyor.
2. Giriş ve çıkışların tümleyenleri alınıyor.

Çıkışı göre karar. (H)



Standard Gate Symbol

Alternate Gate Symbol

$$\overline{x + y} = \bar{x} \cdot \bar{y}$$

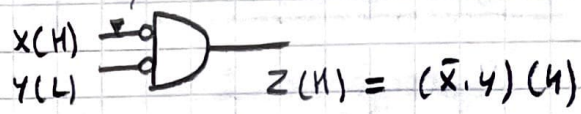


Standard

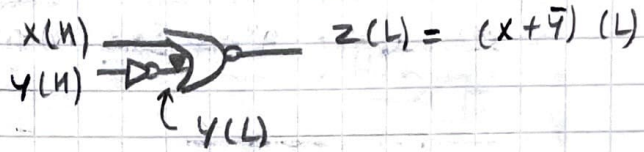
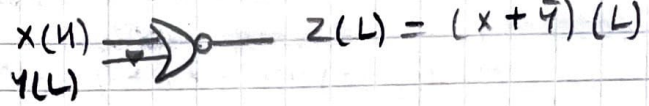
Alternate

LOGIC LEVEL INCOMPATIBILITY

Incompatibility indicator flag



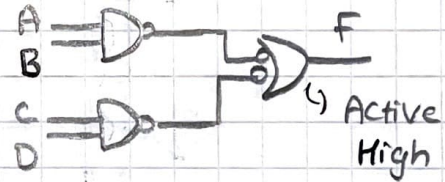
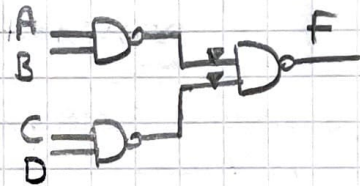
↓ conjugate



ÖRN

Dec	A	B	C	D	F
0	0	0	0	0	0
1	0	0	0	1	0
2	0	0	1	0	0
3	0	0	1	1	1
4	0	1	0	0	0
5	0	1	0	1	0
6	0	1	1	0	0
7	0	1	1	1	1
8	1	0	0	0	0
9	1	0	0	1	0
10	1	0	1	0	0
11	1	0	1	1	1
12	1	1	0	0	1
13	1	1	0	1	1
14	1	1	1	0	1
15	1	1	1	1	1

$n=4$ → L.C. → $m=1$ } Bu önemli



→ Alternatif kapı sembollerini kullanarak lojik uyumsuzluğu kaldırıyoruz.

→ Logic uyumsuzluk nasıl giderilir? Kapıların conjugate'ı kullanılarak

→ Bu 3 devre de birbirinin aynısıdır. Giriş ve çıkışlar aynıdır.

LOGIC FUNCTION

Device Specifications

Logic Function Representation

Reduced or Minimized Form

Implementation

Logic Function Representation

- Tabular Truth Table
- Graphical Karnaugh Map
- Algebraic
- Numeric

LOGIC FUNCTION, TYPES OF LOGIC FUNCTION



} Single L.F.



} General L.F.